321306

12425

3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions : (1) All Questions are *compulsory.*

(2) Illustrate your answers with neat sketches wherever necessary.

(3) Figures to the right indicate full marks.

(4) Assume suitable data, if necessary.

(5) Use of Non-programmable Electronic Pocket Calculator is permissible.

Marks

1. Attempt any FIVE of the following : 10

(a) Define Primary standard and give an example of the same.

(b) Define term Quantitative Analysis.

(c) List various sources of water pollutants.

(d) Name typical properties of water.

(e) Define anode and cathode.

(f) Enlist properties of adhesives.

(g) Write properties of construction chemicals.

2. Attempt any THREE of the following : 12

(a) Explain the concepts of molar solutions.

(b) Describe method of removal of (i) Silica and (ii) Oxygen from water.

(c) Explain Faraday's law of electrolysis with derivation.

(d) Write application of sealants.

3. Attempt any THREE of the following : 12

(a) Explain principle of gravimetric analysis.

(b) Describe phosphate conditioning method of water.

(c) Differentiate : electrochemical cell and electrolytic cell.

(d) Explain the requirements of construction chemicals.

4. Attempt any THREE of the following : 12

(a) Calculate quantity of water to be added to dilute half litre of 2 M sulphuric acid to 0.5 N solution.

(b) Describe municipal water purification in brief.

(c) Write applications of electrochemical series.

(d) State eight applications of adhesives in construction Industries.

(e) Explain safety precautions during application of construction chemicals.

5. Attempt any TWO of the following : 12

(a) Explain complexometric titration with example and reactions. Name the type of indicator used.

(b) Explain with neat sketch deionization process for water conditioning.

(c) Explain application of electrochemistry in various industries.

6. Attempt any TWO of the following : 12

(a) Describe any two properties and applications of under body coatings.

(b) Classify adhesives on the basis of mechanism of adhesion.

(c) Explain any three types of construction chemicals along with its properties and applications.